

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Summary: Senior AI/ML engineer shipping production agentic AI systems, LLM document intelligence, retrieval workflows, validated data pipelines and platforms, and developer tooling for fintech, risk, and compliance teams.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven data-quality anomaly detection for credit trading and risk analytics, reducing false positives by 25% and cutting manual review time so risk teams could focus on critical alerts.
- Developed resilient Python wrapper for KDB+, democratizing data access for 5,000+ users with CI-style tests and optimized Q functions, reducing manual processing by over 10 hours per week.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending application using React, Next.js, TypeScript, MobX, and ethers.js, translating complex wallet and smart-contract flows into secure, performant user workflows.
- Implemented end-to-end CI/CD with GitHub Actions, Husky, Cypress, and Jest to halve release cycles and raise test coverage to 90%, while mentoring a front-end intern to build key UI components.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Engineered and launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in grant funding via a competitive ecosystem program.
- Engineered Python pipelines to extract and normalize a 1M+ Reddit posts/comments dataset and publish a reusable data product for external analytics workflows.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails, Semantic Retrieval

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, Amazon Kinesis, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, SQLite, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Serverless Architecture, AWS Lambda, ECS, EKS, Vercel

PROJECTS

AI Agent Harness Configs - LLM workflow catalog with 40+ reusable skills, MCP-ready configs, and cross-runtime agent automation.

NBA Basketball Database - Validated star-schema warehouse spanning 65K+ NBA games and 13M+ events with DuckDB/SQLite/Parquet exports.

ProxyWhirl - Data-collection toolkit with nine proxy rotation strategies, 100+ sources, and self-healing health checks.

AI/ML Web Feeds - Typed AI/ML source-discovery platform with HTTP/RSS validation, OPML exports, and docs-backed feed workflows.

MCP-Crawl4AI - FastMCP server that lets AI agents crawl, extract, and return structured web content over stdio/HTTP.

MDXPad - Native macOS MDX editor with synchronized preview, multi-provider AI assistance, and Keychain-backed credentials.

Personal Website: w4w.dev - Next.js and MDX publishing system with durable content workflows, SEO, and polished public UI.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering & Operations Research
Aug 2014 - Dec 2020

AWS Certified Cloud Practitioner
Amazon Web Services (AWS) | Nov 2024

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

IBM Data Science Professional Certificate
IBM | Feb 2021